

Development of the Flexible Tactile Sensor for Safe Human-Robot Interaction

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Abstract—With the development of industrialization, more and more robots are added to work with human. However, safety accidents occur frequently in human-robot collaborative scenario. In this article, the flexible tactile sensor is designed to promote the security performance of the collaborative robot. The sensor is made by sponge mixed with graphene and it can convert the pressure of the contact surface into a resistance signal accurately. The sensing unit of the sensor shows the correspondence between the input force and output resistance. In addition, a tactile acquisition system is designed to sample pressure data of the sensor. Results of the human-robot interaction experiments indicate that the proposed flexible tactile sensor responds quickly and it can provide robots with reliable perception capability.

I. INTRODUCTION

As the current industrialization develops rapidly, more and more robots are involved in scenarios that can improve the efficiency of production [1]. However, due to the lack of perception capabilities of current robots, safety accidents occur frequently and threaten the safety of operators seriously [2]. Therefore, how to improve the active safety performance of the robots has become a new research direction [3].

With the development of flexible electronics and functional materials, researchers were trying to use flexible sensors on robots to make robots acquire perception capabilities [4]. When the robot has perception ability, a potential threat to people can be avoided during operation. To enhance the safety of human in human-robot collaboration, Pang et al. [5] proposed the new collaborative robot skin (CoboSkin). The robot skin is made by soft porous materials which is

a type of flexible sensor with the features of variable sensitivity. The reliability of robot skin was verified by robot collision experiments.

Ji et al. [6] designed a flexible tactile sensing array which is based on capacitive mechanism. This flexible tactile sensing array is made by polydimethylsiloxane (PDMS) dielectric and has high sensitivity in a force range of 0–1 N. Therefore, the sensing array can provides robots with the capability to detect small external forces. But the stability and reproducibility of the sensing array was not verified. Kim et al. [7] proposed a flexible dual-type proximity sensor for robots to detect the surroundings. Compared with tactile sensor, the proximity sensor has a longer detection distance. Therefore, when operators approach the robot, the robot can detect operators and make evasive actions. But the anti-interference ability of the proximity sensor is worse than that of the tactile sensor. In order to make the robot perceive touch behavior from a human, Ye et al. [8] designed a novel robot skin with features of high sensitivity and softness. The robot skin was installed on the back of a collaborative robot and the performance of the skin was tested by perceiving four touch modalities. Gbouna et al. [9] proposed the robot skin composed of diamond electrodes and designed the sensor data generation system. In addition, the sensing solution with two operation modes (the interaction mode and the safety mode) was also presented. Pang et al. [10] designed and fabricated a three-dimensional flexible sensor which is made by the piezoresistive nanocomposite. The sensor has good detection performance and could improve the security of the human-robot collaboration. This type of flexible sensor had verified its reliability and performance on the YuMi robot.

To improve the safety of human in human-robot interaction, an flexible tactile sensor is designed in this article. In our previous work [11], a highly accurate flexible sensor was presented to monitor human blood pressure and heart rate. In this work, the sensors of the same material is used to detect pressure of the collaborative robot. The sensor is

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made by sponge mixed with graphene. The flexible shell is prepared by 3D printing mold and can be the buffer between the robot and external collisions. What's more, a tactile acquisition system is designed to sample tactile data of the sensor in this article.

The rest of this article is organized as follows. The structure design of the flexible tactile sensor and the design of the tactile acquisition system is described in Section 2. The characteristic and calibration tests of the sensor are illustrated in Section 3, and the human-robot interaction experiment is carried out to verify the reliability of the flexible tactile sensor. Finally, in Section 4, the conclusion is introduced.

II. MATERIALS AND METHODS

A. Structure Design of the Electronics Skin

One of the key challenges in giving the robots tactile perception ability is how to measure robot's surface pressure accurately. Therefore, it is necessary to design a type of stable and durable flexible tactile sensor to measure pressure. Sponge has the advantages of low cost, high porosity, ultra-light weight and robustness, which is the perfect starting material for the fabrication of various sensors. Graphene is a two-dimensional carbon nanomaterial, which has good electrical conductivity. Because of the two-dimensional structure, graphene is easily adsorbed on organic matter. Therefore, the conductive material which made by sponge and graphene have good characteristic of stability and high sensitivity at the same time. In this article, a graphene/sponge sensor is designed to convert the pressure on the contact surface into a numerical signal accurately. The elastic modulus of ecoflex is closest to that of human skin. Therefore, the ecoflex is selected as the substrate material and shell for the sensing array so as to make the sensor adapt the sensing application scenario of the robot better.

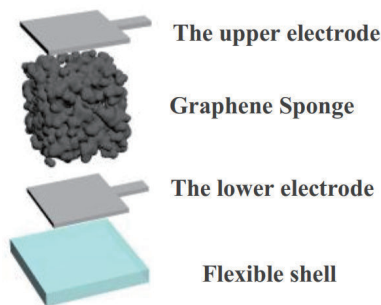


Fig. 1. The schematic diagram of structure of the flexible sensor.

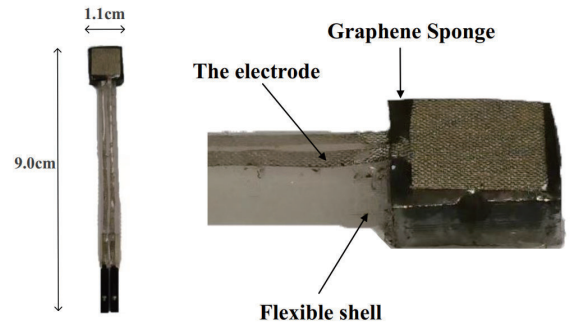


Fig. 2. Photograph of the flexible tactile sensor.

The structure of the flexible tactile sensor is shown in Fig. 1. From bottom to top are the ecoflex flexible shell (substrates), the lower electrode layer, the sensing unit composite layer and the upper electrode layer. The sensing unit is prepared by mixing graphene in the sponge. When pressure is exerted on the sensing unit, the sponge skeleton is compressed and the graphene sheets adsorbed in the sponge skeleton are close to each other. Under the influence of the tunnel effect, more conductive paths are formed, and the resistance of the sensor shows a decreasing trend. When the pressure on the sensing unit is released, the sponge skeleton recovers, the conductive path is broken, then the resistance increases. In addition, the flexible shell of the sensor is also an important part of its function. The 3D printing mold is used to prepare the flexible shell of the sensor, which makes the position of each part in the sensor more accurate. The Fig. 2 exhibit the photograph of the flexible tactile sensor.

B. Design of the Tactile Acquisition System

In order to test the developed flexible tactile sensor, a real-time tactile acquisition system is designed and displayed in Fig. 3. This circuit consists of 5V power, control block chip STM32F103ZET6 (STMicroelectronics, Switzerland) two dual-voltage comparators (LM393, Texas Instrument, USA) and several resistors.

The variable resistance (R_s) represents the flexible tactile sensor. The R_s and the divider resistor R form the voltage dividing circuit, and the output voltage is V_i . The hysteresis control is adopted in the voltage acquisition circuit, which can make the circuit system less sensitive to small changes near comparison voltage and enhance the anti-interference ability of the circuit system. The hysteresis comparators are

composed of the voltage comparator LM393 (U_1 and U_2), R_1 , R_2 , R_3 and R_4 .

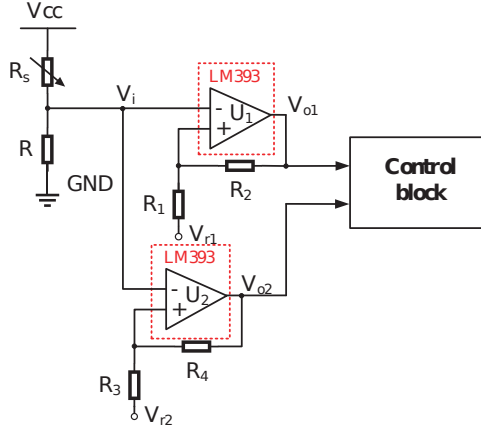


Fig. 3. The schematic of the tactile data acquisition circuit.

V_{r1} and V_{r2} are the reference voltage of voltage comparator U_1 and U_2 respectively. In this circuit, V_{r2} is greater than V_{r1} . When the input voltage is different, the output voltage (V_{o1} , V_{o2}) are also different. When V_i is less than V_{r1} , V_{o1} and V_{o2} are both logic 1. When V_i is greater than V_{r1} but less than V_{r2} , V_{o1} is logic 0 and V_{o2} is logic 1. When V_i is greater than V_{r2} , V_{o1} and V_{o2} become logic 0. The TABLE I indicates the relationship between the input voltage and the output voltage.

TABLE I

THE RELATIONSHIP BETWEEN THE INPUT VOLTAGE AND THE OUTPUT VOLTAGE

V_i	$V_{o1}(\text{logical})$	$V_{o2}(\text{logical})$
$V_i \leq V_{r1}$	1	1
$V_{r1} < V_i \leq V_{r2}$	0	1
$V_{r2} < V_i$	0	0

By using the comparison circuit, the analog signal of the sensor can be convert to digital signal, which can improve the signal processing efficiency and enhance system anti-interference capability. The pressure range can be divided into several intervals easily by the design of the circuit, which will creates conditions for the subsequent security strategies.

III. TEST AND DISCUSSION

A. Calibration of the Sensing Unit

The flexible sensor needs to be calibrated and tested before actual use. The configuration of the test platform is mainly composed of a tension and

compression testing machine (ZQ-990B, Zhiqu Precision Instrument Co., Ltd., Dongguan, China), a digital multimeter, and a computer. The compression testing machine can change test pressure and the measurement accuracy of it is within 1%. The sensing unit is fixed on the surface of the immovable fixture integrated on the testing machine. And the digital multimeter connects the sensing unit with wires to monitor the change of resistance during the test. To prevent the metal fixtures from affecting the test of resistance, the insulation tape is pasted on the surface of metal fixtures. A small plastic block is fastened to the surface of the movable fixture to load force on the sensitive layer on the surface of the sensing unit. When the movable fixture moves downward, the plastic moves with the movable fixture and generate a changeable contact force by press the sensing unit. The contact force is equal to the force exerted by the testing machine. The force exerted by the testing machine was recorded by the force sensor integrated into the testing machine. The raw data of the contact force and resistance are sent to the PC software for further data processing, storage, and analysis. Then, the calibration of the sensing unit of flexible sensor is carried out.

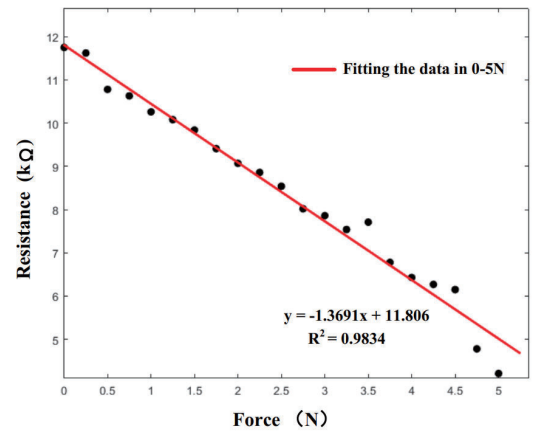


Fig. 4. The calibration of the tactile sensor unit.

The initial resistance (R_0) of the sensing unit is 11.75kΩ. The one-step loading test is designed to represent the sensor's resistance characteristic, and the result are shown in Fig. 4. In this experiment, the pressures range from 0-5N with the step of 0.5N are applied on the sensor. The resistance decreases when a force is exerted to the sensor. It can be seen that there is a nearly linear relationship between resistance value of sensing unit and applied force. The linear

relationship indicates the good performance of the flexible sensor, which can measure small forces on the robot's accurately.

B. Human-robot Interaction Experiment

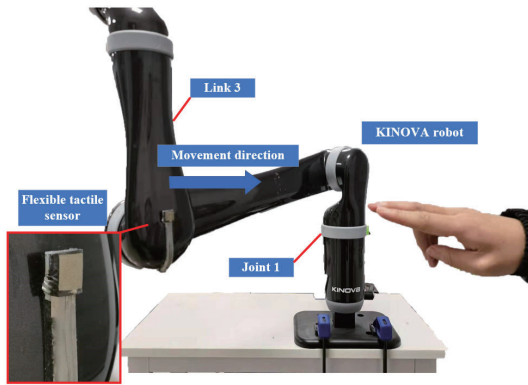


Fig. 5. The human-robot interaction experiment.

In this part, a human-robot interaction experiment is carried out to test the reliability of the flexible tactile sensor. In this experiment, the sensor is integrated with the KINOVA GEN2 collaborative robot (J2N6S200). ROS (The Robot Operating System) is used to receive sensor data and control the robot. The Fig. 5 shows the experimental scene. The joint 1 moves at a constant joint speed ($15^\circ/\text{s}$) with counterclockwise direction. The flexible tactile sensor is attached to the link 3. During the movement of the robot, the operator will touch the sensor with fingers.

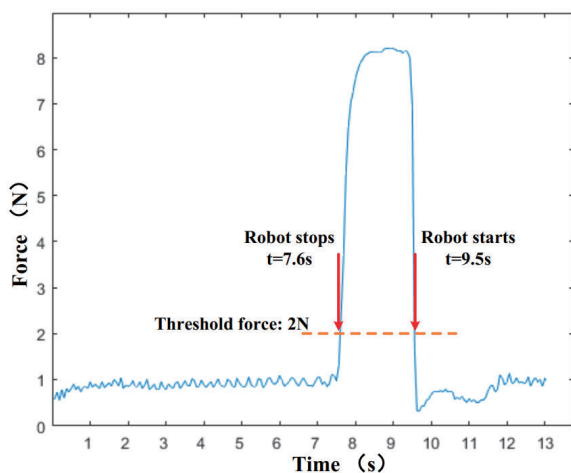


Fig. 6. The tactile data from the sensor during the human-robot interaction experiment.

It can be seen from Fig. 6 that the robot moves normally from 0s to 7.5s, and the sensor data fluctu-

ates around a small value (0.8N). Fluctuations of the sensor data are due to electromagnetic interference inside the robot. At 7.5s, the flexible tactile sensor is pressed by finger, the pressure value rises instantaneously, and the robot stops moving immediately after the tactile data exceeding the threshold force at 7.6s. When the finger keeps pressing the flexible tactile sensor, the pressure value remains high and the robot keeps stopping. When the finger leaves the flexible tactile sensor, the pressure value decreases immediately. At 9.5s, the pressure value is less than the threshold force, and the robot continues to move. The experimental results indicate that the flexible tactile sensor can reliably provide robots with perception capability during the human-robot interaction process.

IV. CONCLUSION

In order to guarantee the safety of human in the human-robot collaboration, this article designed a flexible tactile sensor which can enhance the perception capability of the collaborative robots. The flexible sensor made by sponge mixed with graphene. It can convert the pressure of the contact surface into a resistance signal accurately. The ecoflex is selected as the substrate material for the sensing array so as to make the sensor mimic the softness of human skin better. The 3D printing mold is used to prepare the flexible shell of the sensor, which makes the position of each part in the sensor more accurate. This work also designed the integrated tactile acquisition circuit to collect pressure data in real time. The experiments show that the flexible tactile sensor can detect the force sensitively. What's more, the reliability of the flexible tactile sensor during the human-robot interaction process is also verified in experiments. All the experimental results show that the proposed flexible tactile sensor can provide an efficient approach for secure human-robot interaction.

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